

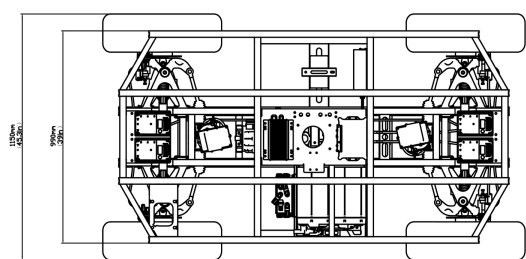
LUCHS B

OFF-ROAD AUTONOMY CHASSIS

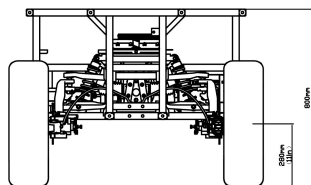
CHASSIS BY RJ TECH LIMITED · AUTONOMY BY TERRAAGENT ROBOTICS



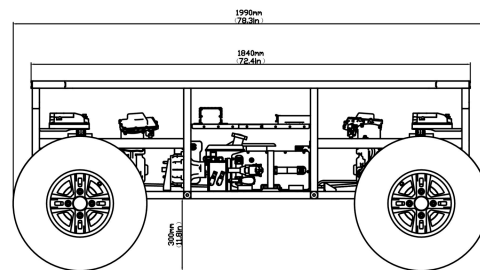
Preliminary v1.4. Specifications may change before release.



TOP



FRONT



SIDE

MODEL LINE-UP – LUCHS A / B / C

	Luchs A compact · most versatile	Luchs B this data sheet	Luchs C custom
Built for	Robots that need many motion modes and interfaces, indoors and out	Off-road and site work: speed, range, clearance, heat and dust	Your specification
External dimensions	800x580x372 mm (A-4WS) 800x665x292 mm (other builds)	1990x1150x800 mm	Customisable
Chassis weight	40–55 kg	230 kg	Customisable
Working / max. payload	— / 80 kg	80 kg / 200 kg	Customisable
Max. climb grade	15°	30°	30°
Ground clearance	77 mm / 220 mm (A-4WS)	300 mm	Customisable
Steering	Skid-steer, Ackermann, dual Ackermann or four-wheel steer	Single or dual Ackermann (dual EPS)	Customisable
Max. speed	11.6 km/h	40 km/h	55 km/h
Drive	4 x 250 W hub motors	4 x in-wheel motors, 5 kW peak each	260 N·m x 4 / 330 N·m x 4
Battery	LiFePO ₄ , 48 V 15 Ah	LiFePO ₄ , 72 V 60 Ah / 73 V 127.4 Ah	LiFePO ₄ , up to 96 V 130 Ah
Endurance	Up to 6 h (no load)	≥60 km (full load), up to 300 km in desert mode	≥60 km, up to 300 km with range extender
Ingress protection	IP33 (IP54 build available)	IP67 (≤440 mm) / IP65 (>440 mm)	IP67
Host interface	CAN 1 Mbit/s (Luchs A protocol v1.1) · RS-485 / RS-232	CAN 1 Mbit/s, LCI v1.0 with heartbeat and live limits	LCI over CAN / CAN FD
Software	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK

SIZE AND WEIGHT

External dimensions (L×W×H) ¹	1990×1150×800 mm
Wheelbase	1430 mm
Track width	970 mm
Ground clearance ²	300 mm
Tyre size ³	22×7–10 in
Chassis weight	230 kg
Working payload	80 kg
Max. payload	200 kg
Suspension travel	50 mm
Approach / departure angle	90° / 90°
Max. side tilt (full load)	30°
Ramp angle	40°

POWERTRAIN

Drive motors	4 × in-wheel motors, 72 V
Rated / peak power per motor	1.8 kW / 5 kW
Peak system power	20 kW
Rated / peak torque per wheel	130 N·m / 330 N·m
Battery	LiFePO ₄ , 72 V 60 Ah (4.3 kWh) or 73 V 127.4 Ah (9.3 kWh)
Battery cycle life	>1000 cycles
Range	≥60 km (full load, 20 km/h, hard dirt) ≥100 km (no load, 40 km/h, paved) Up to 300 km in desert mode
Desert mode (option)	2 battery packs, about 5 kWh each, plus a 1.5 kW range extender built into the chassis.
Payload power	12 V and 24 V outputs, switchable and fused

CHARGING

On-board charger	Yes, full charge in 3 h
Battery swap	<3 min
Power on / off ⁶	Button, remote control or schedule

SPEED AND PERFORMANCE

Drive mode	4WD, 2WD selectable
Steering ⁴	Dual EPS; front Ackermann or dual (front and rear) Ackermann, selectable
Max. road-wheel angle	30°
Steering resolution (wheel end)	0.1°
Min. turning diameter	4.4 m (dual Ackermann)
Max. speed (full load)	40 km/h
Max. acceleration (no load)	0.8 g
Max. climb grade (full load)	30° (58 %)
Max. side slope while driving	40 % (22°)
Service brake	Electro-hydraulic, 240 ms response
Parking brake	Electric parking brake (EPB); holds full load on a 20° slope for ≥5 min, forward and reverse
Vertical obstacle (full load)	150 mm
Trench crossing	400 mm
Wading depth	300 mm
Creep mode ⁵	Yes
Emergency stop	On-board buttons, front and rear bumpers, remote control, host command
Self-test and diagnostics	Yes
Stealth mode ⁷	Yes

AUTONOMY INTERFACE

Protocol	Luchs Chassis Interface (LCI) v1.0: speed and steering-angle commands, live limits, odometry-grade feedback, 200 ms host heartbeat
Physical layer	Dedicated customer CAN 2.0B, 1 Mbit/s, big-endian
Software	ROS 2 (Humble, Jazzy) and ROS 1 (Noetic) drivers, one code base for the Luchs family; C++ SDK, robot model, DBC files, integration guide
Units before LCI firmware	VCU interface V2.6 over CAN, supported by the same software
Chassis IMU	Six-axis, 100 Hz, plus roll and pitch, on the customer CAN
External e-stop loop	Hard-wired loop on the payload connector

ENVIRONMENTAL

Operating temperature	-20 °C to +55 °C
Operating humidity ⁸	95 % (±3 %)
Ingress protection ⁹	IP67 (≤440 mm) / IP65 (>440 mm)
Max. operating altitude	5000 m

COMPLIANCE

Battery transport	UN38.3
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WHAT'S INCLUDED

Chassis	Luchs B, standard configuration
Remote control	Included
On-board charger	Included
Autonomy status light	Included
Documentation and software	User manual, data sheet, LCI and VCU protocols with DBC files, ROS 2 and ROS 1 software with robot model

WARRANTY ¹⁰

Chassis	12 months or 2000 km off-road / 4000 km paved, whichever comes first
Repairs	90 days (parts and labour, same issue)
Spare parts	3 months (genuine parts, authorised channels)

NOTES

1. All values tested by RJ Tech Limited. Modifications may be needed to meet local regulations. The product may differ from the pictures.
2. Measured at standard tyre pressure.
3. Other tyre sizes are available; contact your sales manager.
4. Other steering configurations can be built to order.
5. Creep mode: steady very-low-speed travel, tested on unpaved roads, slopes, mud, sand and rock.
6. Power on and off by button, remote control or a scheduled time.
7. Stealth mode switches off all lights, by remote control or by command.
8. Humidity tested at 20–30 °C.
9. Batteries, motors and connectors mounted below 440 mm are IP67; motor and vehicle controllers are IP65.
10. Extended warranty available. Excludes normal wear, consumables and damage from misuse, accident or unauthorised modification; see full terms.

Pricing and technical consultation: sales@rjtech-offroad.com · www.rjtech-offroad.com